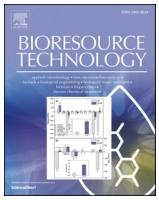
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Unveiling the potential of water as a co-solvent in microwave-assisted delignification of sugarcane bagasse using ternary deep eutectic solvents

Vallari R. Chourasia [a,b], Meena Bisht [c], Kamal K. Pant [a,*], Robert J. Henry [b]

[a] Department of Chemical Engineering, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India
[b] Queensland Alliance for Agriculture and Food Innovation, University of Queensland, Brisbane, QLD 4072, Australia
[c] Department of Chemistry, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India

HIGHLIGHTS

- Aqueous solution of ternary DESs were used for pre-treatment of SB.
- Pre-treatment was performed in 30 mins at 100 °C using microwave heating.
- $MgCl_2$ and $NiCl_2$ were integrated with neutral DESs for pre-treatment.
- 20% aqueous CC:EG:NI achieved 84% delignification and 99% enzyme digestibility.
- EG based DESs showed better pre-treatment performance than GLY based DESs.

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ABSTRACT

Deep eutectic solvents (DESs) have become popular owing to their biodegradability and recyclability. In this study, the influence of water as a co-solvent is demonstrated to enhance the properties of choline based ternary DESs. A fast and energy-efficient microwave-assisted pre-treatment process was developed for delignification of sugarcane bagasse (SB). The effectiveness of SB fractionation was revealed by incorporating Lewis acids ($MgCl_2.6H_20$, $NiCl_2.6H_20$) with the DESs for pre-treatment and Choline chloride: Ethylene glycol: $NiCl_2.6H_20$ (CC:EG:NI) at a molar ratio 1:2:0.016 with 20w% water as a co-solvent provided the most promising result, with 84% delignification and 99% enzyme digestibility. Water was also employed as an anti-solvent to facilitate lignin solubility and exhibited up to 26w% lignin yield from DES liquor with maximum DES recovery of 95% (w/w). Water distinctly affects the density, viscosity, and intermolecular hydrogen bonding of the DES and its impact on the process dynamics is worth further exploration.

1. Introduction

With the advancement of industrialization, rising environmental concerns and diminution of fossil fuel supplies, there has been an ongoing need for alternative renewable resources to meet the energy needs of the world's growing population, and cleaner and greener technologies are a requisite for industrial development. To address this, a biorefinery concept is being introduced, wherein Lignocellulosic biomass (LBM) being the most abundant organic and carbon neutral resource has been potentially used to produce value-added platform chemicals (Patil et al., 2022; Xu et al., 2020). LBM typically comprises of the cellulose and hemicellulose fragments that are sandwiched between the lignin, forming a strong covalently bonded lignin-carbohydrate complex (Asodekar et al., 2021; Chourasia et al., 2021; Hodgson-Kratky et al., 2019; Mankar et al., 2021b). If the biomass is properly depolymerised, the individual fractions can be transformed to a variety of chemicals via chemo-catalytic or bio-catalytic processes and generating fermentable sugars for production of bioethanol and other chemicals. In this context, extensive research has been done, and in the last decade, DESs have made their way into the mainstream of LBM fractionation, owing to their ecologically benign nature and the capacity to selectively extract lignin (Francisco et al., 2012; Pandey & Pandey, 2014). DESs are eutectic mixtures, comprising of a hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD), forming a complex through hydrogen bonding that cause charge delocalization at a specific molar ratio, requiring mild mixing and at temperature of 130 °C or even

* Corresponding author.
E-mail address: kkpant@chemical.iitd.ac.in (K.K. Pant).

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lower (Abbott et al., 2003; Xia et al., 2018). The simplicity of synthesis, tunable physicochemical features, good recyclability, high biodegradability, low cost, and relatively low toxicity are the primary benefits of DESs over traditional Ionic Liquids (Chen & Mu, 2019; Hong et al., 2020). Choline chloride (CC) based DESs with poly alcohol HBDs exhibit significant biocompatibility with enzymes as well as microorganisms due to their nearly neutral pH and are commonly employed as neutral DESs (Chen & Mu, 2019). As a result of bonding between the hydroxyl groups of lignin and HBD, neutral DES pre-treatment has been reported to result in lignin extraction with the maximum β-O-4′ concentration, minimum degree of condensation as well as highest molecular weight, when compared to basic or acidic DES pre-treatment (Dutta et al., 2017; Hong et al., 2020). Francisco et al. (Francisco et al., 2012) first demonstrated that lignin may be readily soluble in DES, while cellulose was sparingly soluble, showing the selectivity of DESs for lignin and negligible solubility of cellulose, thereby displaying effective lignin-cellulose separation after DES pre-treatment. However, recently, studies have revealed that biomass fractionation was not effective with glycerol-based DESs and the efficiency of delignification of polyol-based DESs, mainly CC: glycerol (GLY) and CC: ethylene glycol (CC:EG), could be improved by addition of small quantities of H_2SO_4 or by incorporating Lewis acids like $MgCl_2{\cdot}6H_2O$, $AlCl_3{\cdot}6H_2O$, $CrCl_3{\cdot}6H_2O$, $FeCl_3{\cdot}6H_2O$ into CC:GLY (Chen et al., 2019; Wang et al., 2019). Further, reports illustrated that the saccharification increased by six-fold than the raw biomass while the purity of cellulose reached up to 80.94%, with significant improvement in lignin removal due to active protons and acidic sites provided by the Lewis acids (Chen et al., 2019; Hong et al., 2020; Ji et al., 2021; Wang et al., 2020). Given the diverse processing capabilities of polyol DESs and the crucial function of the active proton, ternary DESs including both a polyol and an inherent acidic component are worth exploring.

Despite substantial advances in DES treatment techniques, these approaches still require a longer time (approximately 6–12 h) to reach the requisite delignification efficiency, which is a major problem in terms of the process's energy efficiency (Kumar et al., 2019). In this respect, microwave heating may be preferable to traditional heating since it occurs at the molecular level and is therefore fast and uniform, reducing the process time from several hours to a few minutes. It was investigated during microwave pre-treatment of pine sawdust using CC: oxalic acid DES, that a substantial lignin yield within an hour was obtained and CC:LA DES with microwave heating, resulted in a lignin yield of 82% in 20 mins for coconut coir biomass (Mankar et al., 2021c; Muley et al., 2019). When comparing the time and energy required for conventional and microwave heating to achieve the same cellulose saccharification efficiency, it was discovered that conductive heating (0.31 kW-h), required more than twice the amount of energy than microwave heating (0.10 kW-h) and 3 h of conductive heating was comparable to 40 mins of microwave heating (Kumar et al., 2019).

Since the physicochemical characteristics of DESs are mainly governed by the interactions of its constituents, temperature change or presence of a co-solvent might significantly alter its fundamental properties (Pandey & Pandey, 2014). Moreover, the fact that several widely accepted DESs are highly hydrophilic in nature, adds to the curiosity of understanding aqueous DES mixtures. Additionally, as previously reported, the use of water as a co-solvent helps reduce the viscosity of the solvents, allowing mass transfer by impacting the rheology of the reaction system and demonstrating better dissolving rates at a lower cost compared to pure DESs (Bisht et al., 2021b). Similarly, water may operate as both a co-solvent and an anti-solvent in the DES pre-treatment process. Water as an anti-solvent can be employed for precipitation of the solubilized lignin, which is the most essential step for separating the solvent from the treated biomass and recovering the DES after processing (Abouelela et al., 2021; Alvarez-Vasco et al., 2016).

In the present comprehensive study, sugarcane bagasse (SB) was pre-treated using microwave heating and by employing aqueous solutions of ternary DESs. India is the second leading manufacturer of sugarcane globally, and has a vital role in the country's economy (Palliprath et al., 2020). Hence, a massive amount of bagasse is produced as a by-product of the sugar industry, most of which goes unutilized. Therefore, this work is an attempt to utilize SB to maximise the production of fermentable sugars by enzymatic saccharification with an environmentally benign pre-treatment process. Using DES-water mixtures as pre-treatment solvents might minimise viscosity as well as the energy inputs and expenses involved with DES recycling, allowing for easier scale-up and downstream processing. See Fig. 1.

2. Materials and methods

2.1. Feedstock and chemicals

SB from an Indian commercial hybrid sugarcane (*Saccharum* spp.) was collected from a local sugar producing mill in New Delhi, India. SB thus obtained was thoroughly washed, sun-dried ground using a laboratory grinder, sieved through a 2 mm screen, processed using ethanol and water in a soxhlet apparatus to eliminate the extractives and thoroughly dried over night to remove moisture (Hames et al., 2008; Sluiter et al., 2005). Finally, the dried biomass obtained was further milled and sieved with the 850 μm sieve size. This SB was stored in sealed containers to keep moisture out until it was used. Choline chloride (≥99%) (CC), glycerol (≥99%) (GLY), ethylene glycol (≥99%) (EG), magnesium chloride hexahydrate ($MgCl_2{\cdot}6H_2O$) (MG), nickel chloride hexahydrate ($NiCl_2{\cdot}6H_2O$) (NI), 4-nitroaniline (NA), N,N-diethyl-4-nitroaniline (DENA), 2,6-diphenyl-4-(2,4,6-triphenyl-N-pyridino)-phenolate (BD 30 - betaine dye 30), ethanol and HPLC grade water were procured from Sigma Aldrich.

2.2. Synthesis of ternary DESs

Four ternary eutectic mixtures were synthesised using CC as a common HBA. CC:GLY:Lewis acid and CC:EG:Lewis acid were synthesised by heating at 80 °C with the molar ratio 1:2:0.016 and stirring continuously to form a transparent and homogenous liquid. Lewis acids used were $MgCl_2{\cdot}6H_2O$ and $NiCl_2{\cdot}6H_2O$ Aqueous mixtures of these DESs were prepared by adding 20% and 40% (w/w) DI water in each of these preparations to get 12 different combinations of the solutions, as summarised in Table 1. Prior to usage, each of them was placed in a vacuum desiccator to avoid capture of moisture from the atmosphere.

2.3. Solvatochromic parameters and viscosity measurement of DESs

Kamlet-Taft procedure was employed to quantitatively measure solvent polarizability/dipolarity (π*), HBD acidity (α) and HBA basicity (β), for all DES combinations using three probes (4NA, DENA, and BD30) (Aryafard et al., 2019). 1 mM concentration of 4NA and DENA while 5 mM for BD were prepared as stock solutions in ethanol. In separate vials, 10 μL of each dye was taken and the ethanol was evaporated using nitrogen gas with high purity, before adding 1 mL of each DES mixture. It was ensured that the sample is thoroughly mixed before noting the maximum absorbance of each sample with the probe (λ_{max}) in a 1 cm^2 quartz cuvette using UV–Vis spectrophotometer (Agilent Cary-4000) between 200 and 800 nm.

The K-T parameters for all DES mixtures were evaluated, and the π*,

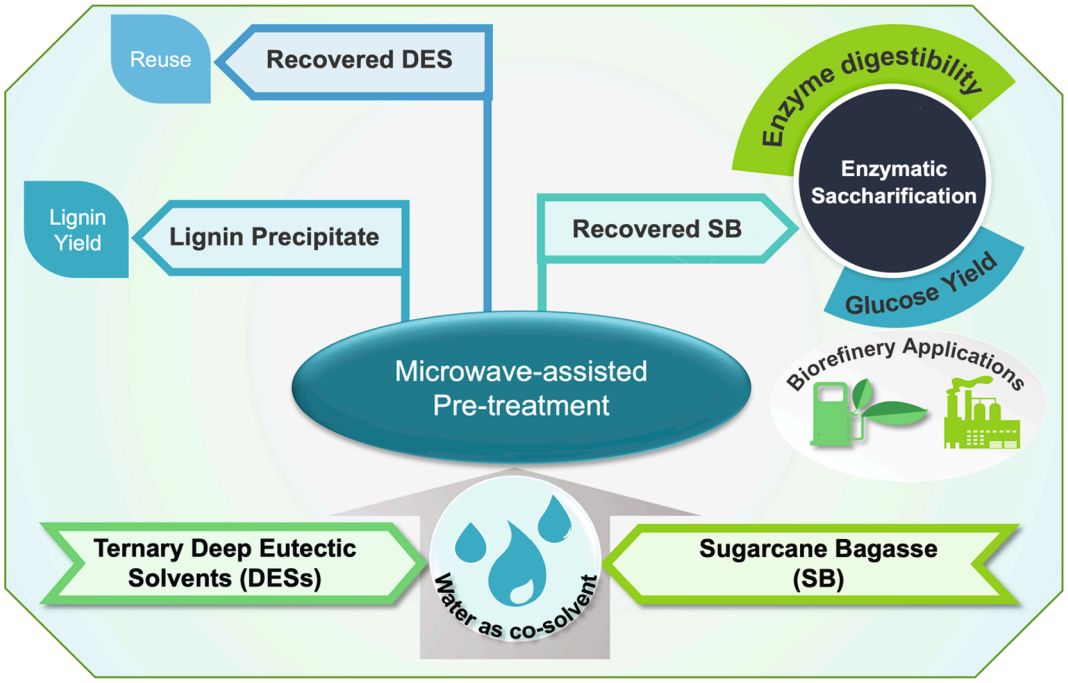


Fig. 1. Schematic representation of the process.

β, and α were calculated from the wavenumbers of maximum absorption of the dyes, DENA (ϑ_{DENA}), 4-NA (ϑ_{NA}), and BD (ϑ_{BD}), respectively, according to the following equations (Aryafard et al., 2019):.

$$\pi^* = 0.427(34.12 - \vartheta_{DENA}$$

$$\beta = \frac{31.10 - 3.14\pi^* - \vartheta_{NA}}{2.79}$$

$$\alpha = 0.186(10.91 - \vartheta_{BD}) - 0.72\pi^*$$

The absorbance maxima of the betaine dye for their minimum intramolecular energy is represented by the molar transition energy (kcal.mol^{-1}) as follows (Aryafard et al., 2019):

$$E_T(30) = 28591.5\vartheta_{BD}$$

$E_T(30)$ is normalized by using tetramethylsilane (TMS) as an extreme nonpolar and water as a highly polar solvent. E_T^N is easier to use as firstly it is a dimensionless factor and secondly it indicates both the polarizability and the HBD ability of the mixture (Rahman & Raynie, 2021).

$$E_T^N = \frac{E_{T(solvent)} - 30.7}{32.4}$$

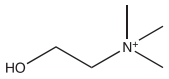
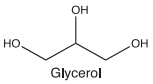
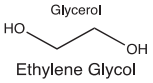
An Anton Paar Rheometer (Model MCR 302) equipped with a parallel plate spindle was used for the viscosity measurement of all the eutectic mixtures. The viscosity was measured at ambient temperature (25 °C) by maintaining a gap of 1 mm between the spindle and the plate, and the flow test was started after a 5 min equilibration time for each sample. Viscosity standard fluids provided by Anton Paar were used to test the rheometer's accuracy and calibration.

2.4. Microwave-assisted DES pre-treatment and DES recovery

In separate experiments, SB was pre-treated with DES mixtures at 10% solid loading with continuous stirring in a microwave reactor (Anton Paar Multiwave PRO). Each reaction was conducted at 100 °C with microwave heating for 30 mins. The slurry was vacuum filtered after the procedure by adding water to the reaction mixture. Since the DES reaction mixtures are viscous, water functions as co-solvent as well as anti-solvent facilitating solid liquid separation along with the precipitation of extracted lignin in the filtrate. The solid residue was dried until it attained a consistent weight in an oven at 60 °C. In this study, the solid dried residue thus obtained was employed as pre-treated biomass.

After DES treatment, the biomass was washed with DI water to separate the DES and biomass. Adding water to the filtrate also facilitates the precipitation of lignin components solubilised along with the DES, retaining the hemicellulose components in the solution (Alvarez-Vasco et al., 2016). The precipitated residue was separated from the filtrate, freeze dried, and weighed for estimation of the lignin yield relative to untreated SB (Liu et al., 2020).

Table 1
List of DESs used for SB pre-treatment[a].

DES	HBA	HBD	Lewis Acid	Mole Ratio (HBA:HBD:Lewis Acid)
CC:GLY:MG	 Choline Chloride	 Glycerol	$MgCl_2{\cdot}6H_2O$	1:2:0.016
CC:GLY:NI			$NiCl_2{\cdot}6H_2O$	1:2:0.016
CC:EG:MG		 Ethylene Glycol	$MgCl_2{\cdot}6H_2O$	1:2:0.016
CC:EG:NI			$NiCl_2{\cdot}6H_2O$	1:2:0.016

[a] Each DES mixture was used in three combinations: Pure DES (100% DES), 80% DES (Pure DES + 20 w% Water) and 60% DES (Pure DES + 40 w% Water).

$$\text{Lignin yield (w\%)} = \frac{\text{Weight of freeze dried residue (g)} \times (\text{AIL of residue (w\%)} + \text{ASL of residue (w\%)})}{\text{AIL of untreated SB (w\%)} + \text{ASL of untreated SB (w\%)}} \times 100$$

The water and DES solution were then heated in a vacuum evaporator to recover the DES by removing the water. The DES recovery was calculated for each reaction as follows:.

$$\text{DES recovery (\%)} = \frac{\text{Intial amount of DES (g)} - \text{Amount of DES obtained after evaporation (g)}}{\text{Initial amount of DES}} \times 100$$

2.5. Enzymatic saccharification

Cellic CTec2 (SAE0020), a commercial cellulase blend was procured from Sigma-Aldrich and employed in this work. The enzyme activity of the enzyme was 119 FPU/mL which was determined using the Filter Paper Assay (Ghose, 1987). To avoid contamination, the enzyme was added in an aseptic environment and enzymatic saccharification was conducted by following NREL LAP (M. G. Resch, 2015), wherein each experiment was performed in separate glass bottles with 20 FPU/g of enzyme loading, biomass, sodium citrate buffer (50 mM, pH-4.8) and 5% sodium azide solution (50 mg/ml in DI water) as an anti-microbial agent. The saccharification was carried out with constant stirring for 72 h at 200 rpm, incubation of 50 °C and an aliquot of hydrolysate was obtained from each reaction every 12 h. The hydrolysate was passed through 0.2 μm syringe filter before transferring to a vial. Using high performance liquid chromatograph (HPLC) (Agilent 1260 Infinity II) equipped with a RI (refractive index) detector and Biorad Aminex HPX-87H ion exchange column, the sugar concentration (g/L) was estimated. All injections were performed in duplicate. The enzyme amenability was calculated using the equation reported in our previous work (Chourasia et al., 2021).

2.6. Compositional analysis

The compositions (%) in terms of carbohydrate content, mainly, glucose, xylose, acid insoluble lignin (AIL), acid soluble lignin (ASL), and ash content of all the pre-treated and untreated biomass were estimated using the NREL standard analytical method (Sluiter et al., 2008). The delignification for all the pre-treated biomass samples was determined using the following the equation (Chourasia et al., 2021):.

$$\text{Delignification}(\%) = 1 - \frac{\text{Total lignin in treated biomass(w\%)}}{\text{Total lignin in untreated biomass(w\%)}} \times \text{solid recovered(g)} \times 100$$

2.7. Physicochemical analysis

FTIR spectroscopy was used to examine the changes in the chemical structure of the biomass and the produced DESs. Thermo Scientific Nicolet iS50 FTIR spectrophotometer with a standard ATR (attenuated total reflectance) mode of operation was utilised to record the FTIR spectra over the range of wavenumbers 4000–500 cm^{-1}.

The diffraction patterns were collected using X-ray diffraction (Rigaku miniflex 600) and the crystallinity index of the untreated and pre-treated biomass were calculated. Cu Kα radiation was created at 40 kV and 15 mA, and the diffraction intensity was measured from 5° to 90° 2θ range, with a step size of 0.02° and a scan speed of 10 s. The crystallinity index (CrI) was calculated as follows:.

$$\text{CrI (\%)} = \frac{\sum \text{Area of Crystalline peaks}}{\sum \text{Area of Crystalline peaks} + \sum \text{Area of amorphous peaks}} \times 100$$

The thermal stability of biomass and synthesized DESs was investigated by TGA (TA instruments, SDT Q600). The thermograms were recorded for each biomass sample weighing approximately 5 mg, while DES mixtures weighing about 10 mg in a closed perforated pan and a heating rate of 10 °C/ min. N_2 was used as a purge gas for complete oxidation from 25 °C to 800 °C at a flow rate of 100 mL/min. The TGA was calibrated using a calcium oxalate standard (greater than99.99%).

3. Result and discussion

3.1. Effect of addition of water on DES properties

3.1.1. Solvatochromic parameters

Table 2 shows the solvatochromic parameters of all DESs estimated using the Kamlet-Taft procedure, and it has been observed that with increasing molar fraction of water (x_w) in DES mixtures, E_T^N also increases, impling an increase in π* and α values of the mixtures. As a result, dipolarity/polarizability (π*) is observed to increase linearly with increase in x_w. This is also in accordance to the previously reported polarizability of conventional binary DESs, namely ethaline (CC:EG), glyceline (CC:GLY) and reline (CC:urea) (Pandey & Pandey, 2014). Thus, the experimental π* values for aqueous DESs, are nearly the same as predicted from the ideal liquid behaviour. However, α values that are determined by the absorbance of the betaine dye, are observed to be slightly lower than the ideal liquids for aqueous DESs. This might be due to the HBD-water bonding, as it dominates over the other bonds which include, HBD-HBD and HBD-chloride ion. This declining behaviour of other hydrogen bonds is observed due to the preferential bonding of HBD to water, because of its high polarity over the other components in the system. In CC:EG and CC:GLY DESs, the amount of hydrogen bonds in HBD-chloride ion pair considerably decrease with addition of more

Table 2
Solvatochromic parameters and Viscosity of DES mixtures.

x_w	CC:GLY:MG				CC:EG:MG				CC:GLY:NI				CC:EG:NI			
	α	β	π^*	$\eta(Pa.s)$	α	β	π^*	$\eta(Pa.s)$	α	β	π^*	η	α	β	π^*	$\eta(Pa.s)$
0.4	0.76	0.60	1.02	0.005	0.68	0.70	1.11	0.004	0.78	0.65	1.02	0.004	0.65	0.71	1.11	0.006
0.2	0.78	0.77	1.08	0.047	0.72	0.82	1.18	0.038	0.93	0.78	1.08	0.043	0.83	0.89	1.19	0.038
0	0.77	0.67	1.06	0.335	0.78	0.71	1.13	0.101	0.82	0.68	1.04	0.314	0.78	0.78	1.14	0.151

x_w = mole fraction of water in DESs.

than 40–50% water, as with high concentration of water in the system, the water moieties get accumulated around the polar molecules of HBD while the alkyl molecules self-associate, causing the HBD-chloride interaction to weaken substantially (Zhekenov et al., 2017).

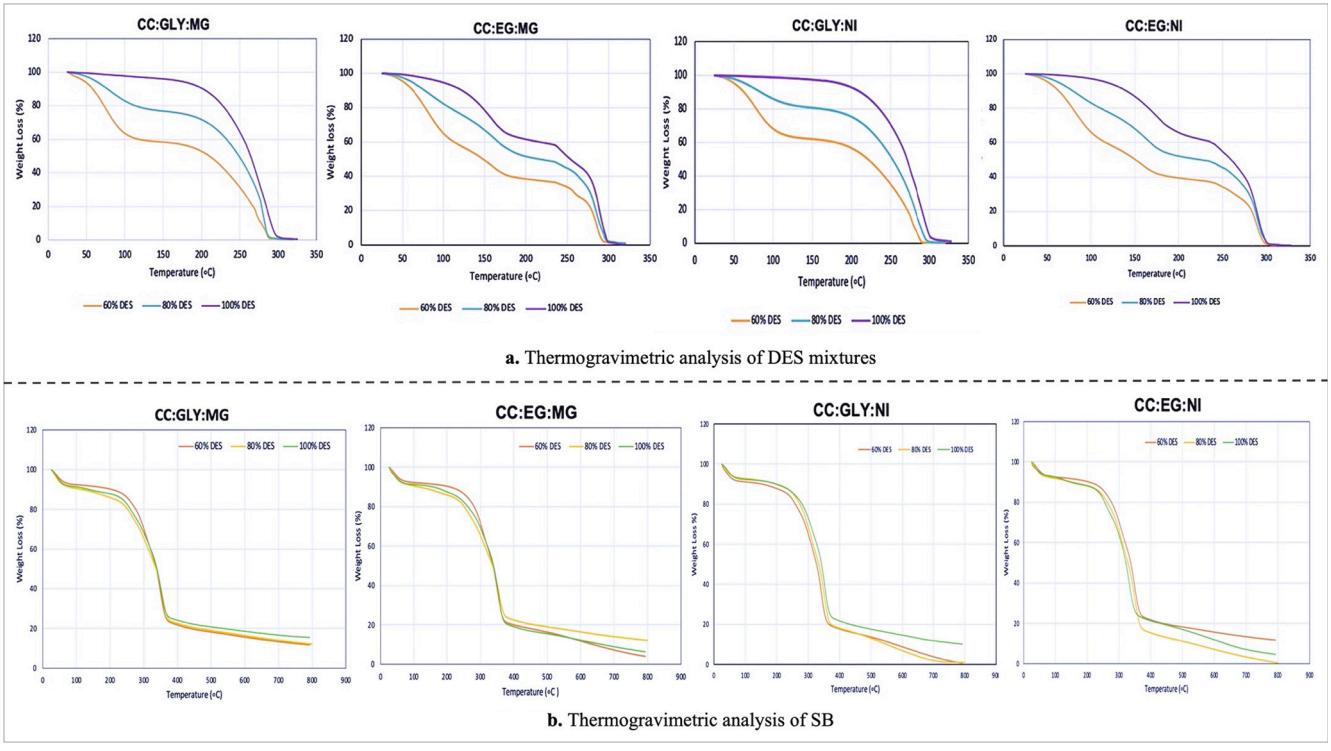
Furthermore, the α and β values correlate with the lignin removal efficiency of the biomass pre-treatment process and DESs that have high α values are expected to readily donate H bonds to release H^+, while those having high β values will exhibit greater ability to accept H-bond (Xia et al., 2018). With higher β value (greater than1), decreased cellulose crystallinity with higher lignin and xylan removal, and enhanced enzymatic digestibility can be observed (Sun et al., 2014). In contrast, compared to the traditional ethaline and glyceline DESs, the acidity and basicity of Lewis acid incorporated ternary DESs is high (Xia et al., 2018). The α and β values for the pure ternary DESs used in this study are higher (Table 2) than the previously reported α and β values for CC:GLY being 0.77 and 0.48 while for CC:EG being 0.89 and 0.55 respectively (Aryafard et al., 2019; Xia et al., 2018).

3.1.2. Viscosity measurements

The viscosity (η) of DES mixtures was estimated by linear regression analysis between shear stress and shear rate at ambient temperature, as listed in Table 2. The viscosity of pure GLY-DESs was observed to be more than 300 times higher than that of water while EG based DESs was nearly 100 times higher. Higher viscosities of some eutectic mixtures are attributed to their strong intermolecular H bond, leading to limited movement of molecules in these mixtures (Gabriele et al., 2019). As the temperature and dilution of the DES mixtures with water rise, the internal resistance of the molecules decrease, leading to the reduction in viscosity (Savi et al., 2019). Additionally, when compared to DESs with lesser viscosity, excessively viscous DESs may obstruct mass transport phenomena during biomass pre-treatment, impacting fractionation efficacy. All the DESs follow Newtonian fluid dynamics, according to the results. Stirring action is frequently necessary in large scale biomass processing to promote efficient mass and heat transfer and the response can be better controlled by employing a Newtonian fluid. In the absence of water, the properties of the HBD are the main influencer of the viscosities of eutectic mixtures. Dilution of DESs with water, on the other hand, caused a substantial reduction in viscosity as the H-bonding interactions between the components gradually weakened (Gabriele et al., 2019; Hong et al., 2020). With the addition of 20w% water, the viscosity value was nearly half of the pure DES, while, depending on the HBD, the initial viscosity was different for all DES combinations, the viscosity of the DESs with 40w% water was almost equal and nearly same as that of water.

3.1.3. Thermal stability

Fig. 2.a displays the thermogram of all DES mixtures, and shows similar thermograms, with minimal displacement. The mass loss began at 25 °C, and the thermograms of all DES combinations displayed no typical peaks of the pure constituents, but instead those contributing to the eutectic mixture, indicating the stability of these DESs and their interaction (Savi et al., 2019). The decomposition temperature of DESs may be defined as the highest temperature at which the DESs remain in their state without undergoing decomposition, and thus it also decides



a. Thermogravimetric analysis of DES mixtures

b. Thermogravimetric analysis of SB

Fig. 2. Thermogravimetric analysis of DES mixtures and SB.

their range of applicability as solvents. The decomposition temperature for DESs is generally between that of their pure components (Delgado-Mellado et al., 2018). As seen in Fig. 2.a, the CC:EG DES mixtures show higher stability when compared to previously reported pure ethylene glycol, which completely vaporizes around 120 °C (Delgado-Mellado et al., 2018). While the complete decomposition of EG DESs occurs in three stages, the first stage of weight loss begins around 80 °C and is attributed to evaporation of EG below its boiling point. The second phase of weight loss that occurs around 200 °C is mainly due to the vaporization of EG when it reaches and exceeds its boiling point. Whereas the third phase of weight loss occurs around 300 °C and is due to CC decomposition. This pattern is also in accordance to the previously reported work (Delgado-Mellado et al., 2018). In contrast, GLY based DESs follow a single-phase degradation pattern with the major mass loss caused by the simultaneous glycerol vaporization and CC decomposition which occurs around 300 °C as the boiling point of glycerol, is almost same as the decomposition temperature of CC. Thus, the thermograms indicate that DESs formed using HBD with low volatility like glycerol show higher thermal stability than EG based DESs.

3.1.4. FTIR analysis

FTIR spectra for all the DES mixtures showed similar spectra for both GLY and EG DESs. Almost same vibrational bands are visible in the DESs due the structural similarities between EG and GLY molecules. Major absorption band is observed at 3550–3300 cm^{-1} and 550 cm^{-1} depicting the O–H stretching and O–H bending of hydroxyl groups, respectively. While 2950 cm^{-1} band is for C–H stretching, and 1330 cm^{-1} represents C–H scissor and bending. Whereas 1505 cm^{-1} shows the CH_2 bending of mainly alkyl groups, and 1140 cm^{-1}, 1035 cm^{-1}, and 867 cm^{-1} refer to alcoholic functional groups with C-O stretching, C–C-O symmetric and asymmetric stretching respectively. Additionally, the 1110 cm^{-1} and 3040 cm^{-1} bands are due to the C-O–H bending and N–H stretching, respectively. Stretching of ammonium C-N group of choline chloride is represented by the vibrational band at 960 cm^{-1}, which is evident only in DESs (Gabriele et al., 2019). To explore the effect of addition of water on the chemistry of the DES mixtures, it was discovered that the strong intermolecular hydrogen bonding between the HBA and HBD of DESs are weakened, causing the O–H band to intensify or move to higher frequencies as the water content in the DESs increased (Gabriele et al., 2019). Because glycerol contains three alkyl O–H groups it induced more molecular contraction in aqueous mixtures of CC:GLY DES, as previously reported, polyol-based DESs displayed superior H-bonding between water and its HBD than aqueous combinations of typical CC: Urea DESs (Pandey & Pandey, 2014).

3.2. Impact of water as a co-solvent for SB fractionation

3.2.1. Compositional analysis

Compositional analysis was performed to determine the structural components of SB before and after DES pre-treatment. Carbohydrate content, mainly in the form of glucose and xylose was estimated, along with total lignin (ASL + AIL), and total ash content (Chourasia et al., 2021; Patil et al., 2020; Sluiter et al., 2008). Interestingly, the ternary DES pre-treatment resulted in overall lignin removal of around 77% on an average, CC:EG:NI with 20w% water showing the highest delignification of 84% (Fig. 3). Overall, the DES pre-treatment was found to be less effective on hemicellulose, as there was no significant improvement in the xylan content after pre-treatment. According to Abbott et al. (Abbott et al., 2006), DESs exhibit specificity towards lignin extraction by stabilising the cellulose complex, because of strong H bond interaction with the CC hydroxyl groups, making the solubility of cellulose or hemicellulose difficult. Nonetheless, in this study, the glucan content,



Fig. 3. Compositional analysis of untreated and pre-treated SB and Delignification (%).

increased from 45% in untreated SB to approximately 63% for the pre-treated SB. The increase in glucose content following DES pre-treatment might be attributed to the enhanced total cellulose availability after the fractionation of LBM and alteration of the crystalline cellulose, as revealed by the XRD analysis. Furthermore, the addition of 20w% water during DES pre-treatment consistently improved total lignin extraction for all DES mixes, but the addition of 40w% water decreased fractionation efficiency. This is consistent with the previously published study by Kumar et al. (Kumar et al., 2016), who found that increasing the water content by more than 5% influenced lignin extraction, whereas 50% water extracted the least lignin. Furthermore, when the amount of water added to the DESs was increased, the viscosity of the DES decreased (Table 2). Because water has lower density and viscosity than pure DES, with increase in the concentration of water in DESs, the density and viscosity of the solvent mixture tend to decrease. This also explains why EG based DES mixtures with lower viscosity remove more lignin than GLY DESs (Fig. 3). In contrast, as expected, solid recovery was greatest after DES pre-treatment with 40w% water and least for the pure DESs. Given that typical DESs with near neutral pH, such as those investigated in this work, demonstrate less efficient biomass fractionation when compared to acidic DESs, the pre-treatment performance of DESs has been significantly improved with the addition of Lewis acids (Chourasia et al., 2021; Hong et al., 2020; Wang et al., 2020). The metal chloride has dual-acid characteristics, that appreciably improve the catalytic conversion process during biomass pre-treatment (Ji et al., 2021). Additionally, it has been hypothesised that after incorporating Lewis acid into the ternary DESs, to promote increased depolymerisation of biomass and lignin removal, the hydrogen bonds create a supramolecular complex with the protons generated by the partial hydrolysis of Lewis acid (Ji et al., 2021; Wang et al., 2019). At the same time, the pH of Lewis acid is critical in biomass fractionation because more acidic Lewis acids interact more with biomass to trigger a dehydration process that breaks the lignin-carbohydrate complex (Ji et al., 2021). This might explain why lignin removal was marginally greater (Fig. 3) after pre-treatment with DESs with $NiCl_2$ (pH = 4.0–6.0) compared to lignin removal using DESs with $MgCl_2$ (pH = 4.5–7.0).

3.2.2. XRD analysis

The crystallinity index (CrI) of untreated and pre-treated SB was examined using XRD analysis in this study to assess the performance of DES pre-treatment. XRD diffractograms demonstrated 2 major peaks at 22.49° and 16.8°, representing amorphous and crystalline regions respectively. Two main parameters might dictate the mechanism behind the CrI changes during pre-treatment process: swelling and dissolution of cellulose (decreases CrI), removal of the amorphous lignin and hemicelluloses (increases CrI) (Chourasia et al., 2021). CrI (%) of untreated SB was found to be 56.8%. SB pre-treatment using 60w% DESs demonstrated a slight decrease in the CrI to 53.5%, 50.6%, 52.7% and 47.5% for CC:GLY:MG, CC:EG:MG, CC:GLY:NI and CC:EG:NI respectively. This may be attributed to the dissolution of the crystalline cellulose, producing more amorphous cellulose available for enzyme hydrolysis which is also observed in case of conventional CC:GLY and CC:EG DES pre-treatment of SB (Ci et al., 2020; Li et al., 2021). In contrast, the CrI for pure DESs (100w% DESs) increased to 61.8%, 63.7%, 59.9% and 64.1% for CC:GLY:MG, CC:EG:MG, CC:GLY:NI and CC:EG:NI respectively. This clearly demonstrates the ability of pure DESs to remove lignin while leaving the cellulose structure reasonably intact for subsequent enzyme hydrolysis. These findings, which were consistent with the chemical compositional analysis, suggested that amorphous lignin and hemicelluloses were eliminated to varied degrees. This is also consistent with the previously reported results, where addition of a series of Lewis acids to the DESs gave CrI values ranging from 59.6% to 70.8% for Hybrid Pennisetum pre-treatment (Wang et al., 2020). Further, SB pre-treated with 80w% DES mixture showed no significant change in the CrI which most likely can be due to the balance of both these parameters.

3.2.3. FTIR analysis

FTIR analysis indicates the chemical structure modifications of pre-treated and untreated biomass. Although there were no significant changes seen in the functional groups of cellulose after pre-treatment, there were considerable alterations in the distinctive peaks related to lignin. The FTIR spectra of all samples revealed a wide band at 3200–3500 cm^{-1}, which is indicative of the O–H stretching of the hydroxyl groups. This band is often deeper in cellulose owing to intermolecular hydrogen bonding, but it is broader in lignin due to phenol hydroxyl groups (Mamilla et al., 2019). The spectra showed C–H stretching vibration around 2910 cm^{-1}, which might occur due to the deformation of the methyl and methylene groups of the cellulosic materials (Kumar et al., 2016). Diminishing of the shoulder peak characteristic of lignin at 1750 cm^{-1} after DES pre-treatment clearly indicated the depolymerization of the biomass. Absorbance peaks around 1653–1631 cm^{-1} resulted from O-H bending of adsorbed water which is observed to increase in intensity with the addition of water in the DESs (Kumar et al., 2016). Another important absorption band is at around 905 cm^{-1} which is associated with the β-glycosidic linkages between glucose residues (Bisht et al., 2021a). The vibrational bands for C–C and C-O stretching representing aromatic ring vibrations of lignin are observed at 1200 and 1250 cm^{-1} and are seen to decrease in intensity indicating the lowering of lignin content from the biomass (Mamilla et al., 2019).

3.2.4. Thermogravimetric analysis

The thermal stability of untreated and pre-treated SB was further compared using TGA thermograms. For all the samples, similar patterns were observed as the thermal degradations occurred mainly in three phases. In the first phase, dehydration occurred from 25 °C to 210 °C where primarily the moisture, extractives or volatile compounds were removed from the biomass. In the second phase, major degradation of the biomass occurred from 210 °C to 400 °C. Degradation of cellulose, hemicellulose and aliphatic compounds occurred in this phase (Sahoo et al., 2021). The decomposition temperature (T_d) can be estimated from this phase, and it was observed from the thermograms that T_d for untreated biomass (~395 °C) was slightly higher than the pre-treated biomass (~385 °C) (Bisht et al., 2021a). Finally, the third stage occurs above 400 °C, where lignin decomposition occurs, and the biomass can be further converted to char.

3.3. Lignin precipitation and DES recovery

Anti-solvents such as water, alcohol, or water-alcohol mixtures, are widely used for washing the biomass after DES or IL processing, to

Table 3
Lignin yield (%) and DES recovery (%).

	DES Recovery (%)	Total lignin (AIL + ASL)(%)	Lignin Yield (%)
Untreated SB	–	18.60	–
100% CC:GLY:MG	78.85	24.50	10.49
80% CC:GLY:MG	89.00	32.90	16.45
60% CC:GLY:MG	94.52	25.67	13.30
100% CC:EG:MG	94.46	39.53	16.58
80% CC:EG:MG	95.59	40.36	24.01
60% CC:EG:MG	94.16	40.23	21.48
100% CC:GLY:NI	82.46	30.55	10.52
80% CC:GLY:NI	94.50	32.08	14.45
60% CC:GLY:NI	95.80	31.92	12.36
100% CC:EG:NI	90.99	40.96	21.33
80% CC:EG:NI	95.87	43.26	26.05
60% CC:EG:NI	95.71	42.37	22.87

facilitate the recovery of solvent and dissolved lignin (Abouelela et al., 2021; Alvarez-Vasco et al., 2016; Bisht et al., 2021b; Francisco et al., 2012). In this study, the potential of water as an anti-solvent was explored.

As expected, the amount of lignin precipitated in aqueous DESs was higher than the pure DESs and the addition water required to precipitate the same amount of lignin was also significantly less for aqueous mixtures of DESs. The yield of lignin precipitated ranged from 10.5 to 26.4% depending on the viscosity of HBD and the dilution of water in the DES pre-treatment (Table 3). For every gram of pure DES or DES-water mixture 3 g of water was added to wash the biomass (Abouelela et al., 2021). Thus, using aqueous DES mixtures for pre-treatment proves to be beneficial as the amount of water required for washing substantially reduces thereby improving the overall energy efficiency and economics of the process. It is noteworthy that the total lignin (AIL + ASL) recovery yields were found to be higher than the initial lignin content of untreated SB (18.6%), which suggests the presence of impurities in the precipitated lignin. Additionally, the compositional analysis of the precipitated lignin indicated that a substantial amount of DES may have remained in the residue, as suggested by the presence of glycerol and ethylene glycol in the HPLC analysis. This is also consistent with the findings of Alvarez-Vasco et al. (Alvarez-Vasco et al., 2016), who discovered that when poplar was treated with CC:Lactic acid, less than 60% lignin purity was obtained and more than 30% DES in terms of lactic acid was detected in the precipitated lignin compositional analysis. Moreover, the carbohydrate (glucan and xylan) content in the precipitated lignin was less than 5%.

However, it has been reported, that although with water in DESs, separation of lignin is facilitated, but it also caused DES co-precipitation with lignin (Alvarez-Vasco et al., 2016). Hence, to avoid the loss of DES during co-precipitation, other solvents or solvent–water mixtures can be explored as anti-solvents for DESs.

The evaluation of DES recycling is critical for designing a cost-effective and sustainable process based on green chemistry principles. Several reports have shown, DESs can be easily recycled from the reaction mixture by vacuum filtration or freeze drying (Ji et al., 2021; Mamilla et al., 2019; Mankar et al., 2021a; Wang et al., 2020). In this work, DESs were recovered using vacuum evaporation of the dissolved water. On average, the recovery of DES was found to be 86.7%, 93.7% and 95.1% for pure, 20w% aqueous and 40w% respectively (Table 3). The recyclability of DESs is expected to be dependent on the nature of HBD, extent of dilution of water in the DES mixtures and efficiency of biomass washing. In this regard, the recovery of DESs with EG was slightly higher than the DESs with GLY as the HBD. In previous studies, the activity of DESs is reported to be maintained until four cycles and slight decrease in the fifth cycle (Ji et al., 2021; Mamilla et al., 2019; Mankar et al., 2021a). In this work, since the novel CC:EG:NI DES demonstrated the best results overall, the recyclability of this DES was studied up to three cycles. After the third cycle, the glucose content of the pre-treated SB slightly decreased from 66.5% to 49.8% for pure DES, while the delignification decreased from 84% to 76.2%. Changes in the chemical characteristics and loss of DES during the process might be the primary causes of the decline in performance.

3.4. Enzyme digestibility measurement

The ultimate goal of any pre-treatment procedure in a bio-refinery idea is to improve the valorization of the lignocellulosic biomass to create fermentable sugars that can then be converted to bio-ethanol. Cellulase used in this study, is a complex enzyme, consisting of β-glucosidase, *endo*-β-glucanase, and *exo*-β-glucanase along with a highly active xylanase (Ci et al., 2020). Therefore, this enzyme can contribute in the enzymatic saccharification of both cellulose and hemicellulose. Enzymatic digestibility of the untreated and DES pre-treated SB at different time intervals is displayed in Fig. 4. DES pre-treatment of SB facilitates the enzymatic accessibility of cellulose by

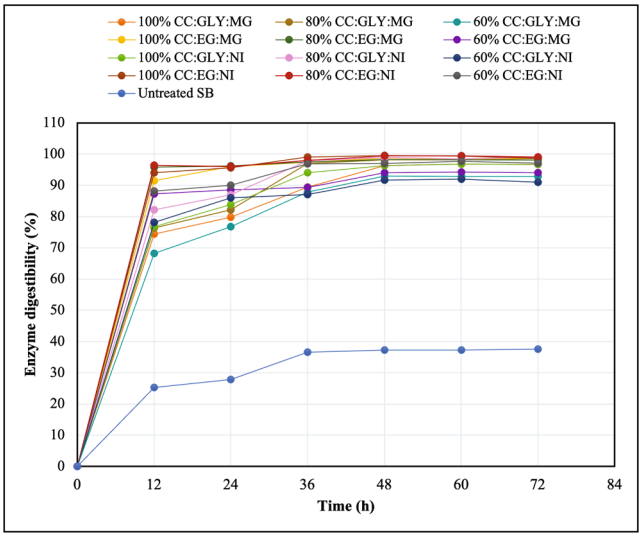


Fig. 4. Enzymatic digestibility (%) after 72 h for untreated and DES pre-treated SB.

fractionation of biomass thereby increasing the solubility of lignin and hemicellulose (Ling et al., 2020; Ostby et al., 2020). Therefore, enzymatic saccharification enhances after pre-treatment compared to the untreated SB (Fig. 4). Moreover, the glucose conversion of SB increases up to 48 h, but maximum glucose conversion with high hydrolysis rate occurs in 12 h revealed by a drastic elevation in the enzyme digestibility from 0 to 12 h. However, after 48 h the hydrolysis rate decreases and minor change is observed in the enzyme digestibility causing almost a plateau region, which is mainly due to the saturation of enzyme active sites and enzyme inhibition due to glucose or cellobiose (Andric et al., 2010). Enzyme digestibility was seen to be only 37.5% after 72 h while, evidently, the DES-treated SB demonstrated higher enzymatic digestibility up to 99%. In all the experiments, the digestibility is observed to increase up to 48 h and then plateau up to 72 h, indicating 48 h to be the optimum time for DES pre-treatment. Further, as seen from the compositional analysis, lignin removal is directly correlated with the glucose concentration after hydrolysis and thereby the enzyme digestibility. Since the CC:EG:NI treatment removed the most lignin (84%) it also had the greatest enzyme digestibility (99.4%). However, no strong correlation was seen between the xylan concentration and the enzyme digestibility. Additionally, a positive linear correlation was also observed between the β values and the glucose concentration after hydrolysis. As a result, β values might be utilised as a critical parameter to predict the pre-treatment efficiency of DES mixtures. As a result, this study indicated more than 70% cellulose conversion to glucose, with the highest reaching 99% with a DES pre-treatment of 30 mins. The incorporation of Lewis acids as HBDs to form novel ternary DESs provided a fairly strong acidity to the conventional neutral DESs, thus enhancing its pre-treatment performance. Given the total process dynamics, the high performance of the CC:EG:NI DES pre-treatment, which achieved 99% enzyme saccharification in 48 h, might be tremendously useful in a bio-refinery concept, as it saves 12 h relative to the traditional process. However, in order to attain the intended pre-treatment outcome, it is critical to understand the precise goal of pre-treatment, whether for dissolution or hydrolysis enhancement, so that the appropriate solvent could be employed.

4. Conclusion

Water as a co-solvent improved the pre-treatment efficiency of DES pre-treatment, resulting in 84% delignification and 99% digestibility of SB. Although 60w% DESs aqueous mixture performed poorly, 80w% DES aqueous mixture surpassed pure DESs for thermal stability,

crystallinity, delignification, and enzyme digestibility. Additionally, 95% (w/w) DES recovery was achieved using water for lignin precipitation. The viscosity, thermal stability, and solvatochromic parameters revealed, EG DESs are superior solvents to GLY DESs. Integrating MG and NI to neutral DESs provided acidity to enhance their pre-treatment efficiency. Microwave heating augmented the process dynamics by saving time and energy during the DES pre-treatment process.

CRediT authorship contribution statement

Vallari R. Chourasia: Conceptualization, Methodology, Investigation, Writing – original draft. **Meena Bisht:** Conceptualization, Methodology, Writing – review & editing. **Kamal K. Pant:** Supervision, Resources, Funding acquisition, Writing – review & editing. **Robert J. Henry:** Supervision, Resources, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2022.127005.

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